Oliver Zhang

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Summary

I'm a versatile mechanical engineer with extensive experience designing and building complex electromechanical systems. I thrive in highly collaborative interdisciplinary teams, and I love finding elegant solutions that bring innovative ideas to life.

I'm an Australian citizen.

Education

Carnegie Mellon University, BS in Mechanical Engineering with an additional major in Robotics with University Honors – Pittsburgh, USA — Aug 2015 – May 2019

- Completed a wide range of course work from mechanical engineering fundamentals to computer vision, artificial intelligence, and systems engineering.
- Worked with cross-disciplinary teams on multiple projects, including building an amphibious vehicle, a self-navigating drone, and an autonomous delivery robot.

Experience

Senior Mechanical Engineer, PA Consulting – San Francisco, USA — July 2019 – present

Collaborating with diverse engineers and designers to find the path through complex challenges and turn inventive ideas into innovative products.

- Delivering projects across many different industries, from medical robotics to industrial food production.
- Working through the entire product development process, from concept to manufacturing.
- Collaborating with a wide range of clients from agile tech startups to established global leaders.
- Brainstorming ideas and developing concepts to find the best path forward.
- Building prototypes at the right fidelity to rapidly drive to the optimal design.
- Architecting complex systems by breaking them down into key interactions and applying first principles analysis and modelling.
- Coordinating highly interdisciplinary teams spread across multiple geographies, with electrical engineers, software engineers, and industrial designers.
- Growing the team's capabilities through weekly forums and training sessions, covering topics like analysis, sketching, or programming.

Project Engineer Intern, Autel Robotics – Pittsburgh, USA — May 2017 – Aug 2017

Worked in a team of three to implement tracking and gesture recognition systems for a quadrotor. Developed programs using MATLAB and cross compiled using C/C++ for embedded execution.

Skills

Programming: Python, C/C++, MATLAB, ROS, Unity

Mechanical Engineering: Technology development and prototyping, Analysis of complex systems, Electromechanical systems, Design for manufacture, SolidWorks, GD&T, Tolerance stack analysis

Systems Engineering: Requirements development, System architecture design, Risk management

Languages: English (native speaker), Mandarin (native speaker)

Patents

US11058204B2 — July 2021

Automated total nail care systems, devices, and methods

Select Projects

Novel Electromechanical Machine Aug 2025 – Feb 2026

Developing a consumer facing electromechanical machine for a global technology company.

- Owned design of an Alpha prototype system with 7 actuated degrees of freedom.
- Designed complex motion systems in heavily space constrained and finish constrained environment.
- Interfaced with a diverse team of electrical engineers, software engineers, and industrial designers.
- Led prototype build effort, coordinating integration of Beckhoff PLC electronics, wiring of motors and sensors, and assembly of complex mechanical subsystems.
- Final prototype's reliability and quality exceeded expectations and allowed client to fully de-risk intended user experience with company executives.

Novel Cable Splicing Machine Aug 2024 – Aug 2025

Developing an underground cable splicing machine that automates a complex process typically done by a trained technician.

- Built complex system requirements drawn from multiple stakeholders, and managed key risks.
- Optimised automation workflow to improve cycle time and reduce manual intervention.
- Designed system architecture to maximise reliability and meet stringent weight and space constraints.
- Developed multiple electromechanical mechanisms that each perform dexterous operations that previously required highly coordinated human actions.

Novel Smart Fitness Machine Aug 2022 – Aug 2024

Developed a novel fitness machine that is controllable wirelessly and can report data from multiple integrated sensors.

- Implemented and refined controls algorithms to realistically simulate inertia with an electric motor.
- Built dynamometer rig with software that simulates human motion and displays metrics in real-time.
- Developed system requirements, managing traceability and designing test plans.
- Led verification and validation of the prototypes across multiple locations globally.
- Coordinated a highly cross-functional team with industrial designers, electrical engineers, firmware engineers, and mechanical engineers.
- Delivered high quality prototypes that demonstrated best-in-class user experience and performance, standing up to rigorous testing by world-class athletes.

Optimising Industrial Food Production Aug 2023 – Nov 2023

Analysed an industrial food production line to identify root causes of defects and improve throughput without changing line footprint.

- Analytically modelled complex dynamic behavior of key components in the production line leveraging high-speed camera footage collected on-site.
- Developed several concepts for improvement, covering the full solution space.
- Client implemented designs and achieved 47% increased throughput without increasing line footprint.

Novel Subsystem Prototype Sept 2020 – Apr 2021

Developed a proof-of-principle prototype to de-risk a key robotic subsystem.

- Developed a complex analytical model of the behavior of elastic materials fed from a spool.
- Experimented with medical grade extruded tube materials, coatings, and cross-sectional profiles to generate desired properties such as bending stiffness, resilience, creep, and friction.
- Designed and built a robotic system that process various sensor inputs to drive a custom control algorithm. The system uses clinical trial data to simulate real user motion, allowing high fidelity testing of expected behavior.